Is the Gender Pay Gap Largest at the Top?

by

Ariel J. Binder
U.S. Census Bureau

Amanda Eng
U.S. Census Bureau

Kendall Houghton
U.S. Census Bureau

Andrew Foote
U.S. Census Bureau

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Abstract

No: it is at least as large at bottom percentiles of the earnings distribution. Conditional quantile regressions reveal that while the gap at top percentiles is largest among the most-educated, the gap at bottom percentiles is largest among the least-educated. Gender differences in labor supply create more pay inequality among the least-educated than they do among the most-educated. The pay gap has declined throughout the distribution since 2006, but it declined more for the most-educated women. Current economics-of-gender research focuses heavily on the top end; equal emphasis should be placed on mechanisms driving gender inequality for noncollege-educated workers.

Keyword*:* gender pay gap, education, conditional quantile regression, glass ceiling, labor supply

JEL Classification: I24, J16, J31

* U.S. Census Bureau, Center for Economic Studies. Any opinions and conclusions expressed herein are those of the authors and do not reflect the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7527717: CBDRB-FY2024-CES010-003.) Corresponding author: Ariel Binder, ariel.j.binder@census.gov.

After declining for multiple decades, the mean difference in pay between women and men in the United States changed little in the late 1990s and early 2000s (Blau and Kahn, 2006). Concerns about slowed progress motivated Lisa Belkin's famous "Opt-Out Revolution" news story as well as Claudia Goldin's (2006) discussion of whether women had reached their "natural rate" of involvement in the workforce. Researchers began to document a "glass-ceiling" pattern: among full-time, full-year (FTFY) workers, pay differences were largest among women and men at the tops of their respective pay distributions (e.g. Albrecht et al., 2003; Arulampalam et al., 2007; see also Blau and Kahn, 2017). Noonan et al. (2005) estimated a large and unchanging gender pay gap across two cohorts of lawyers; Bertrand et al. (2010) found divergent career paths following graduation that cumulated to a very large pay gap among elite MBA graduates.

What has happened to the U.S. gender gap since that time and how does it currently vary across earnings and skill positions in the labor market? While recent literature has continued to study select samples at the top of the distribution (e.g. Fernandez and Campero, 2017; Cortés et al. 2021; Murciano-Goroff, 2022; Wasserman, 2023; Foster et al., 2023), less attention has been paid to these broader, population-level questions. We begin to answer them in this short paper, using representative data on annual earnings from the American Community Survey. Relative to the literature, we consider more-recent data and include non-FTFY workers in the sample, both of which we show affect estimates of the pay gap.

While the top-end focus of the recent literature is not unjustified—the pay gap remains particularly large at the top of the pay distribution—we show that the gap is at least as large at the bottom of the pay distribution. The mean gap among high school dropouts slightly exceeds the gap among advanced degree holders; this is driven by a large gap toward the bottom of the distribution among dropouts that more than offsets the large gap toward the top of the distribution among

advanced graduates. We show that labor supply differences create more pay inequality at the mean among dropouts and high school graduates than they do among advanced graduates.

Finally, in contrast to the perception that there has been little or no progress toward gender inequality in the twenty-first century, we show that the gender pay gap has declined on average and throughout the education distribution since 2006. However, the gap declined substantially more among bachelor's graduates and advanced graduates than it did among other education groups. Concurrently, the gender gap in the share of individuals working FTFY declined much more among these groups than it did among the less-educated, which we connect to the pay gap changes via simple decomposition methods.

I. Data

We construct our sample from the 2018-2022 American Community Survey (ACS) 1-year files, excluding the COVID-19 pandemic years of 2020 and 2021. We use the restricted-access file versions.[1] Our sample consists of prime-working-age individuals (ages 25-54) who were not enrolled in school in the three months prior to the survey; were employed for at least one week out of the prior year; and who earned at least 100 times the federal minimum wage during the prior year.[2] It contains approximately 4.04 million individuals.

This sample includes individuals who did not work FTFY, consistent with our goal of estimating the full earnings distribution and how it varies by gender. We consider how the exclusion of these individuals affects our results using the subsample of individuals who worked

[1] These files contain larger sample sizes than the public-use versions and no editing of tail earnings observations, so they are well-suited to estimating finer moments of the pay distribution.

[2] Apart from the bottom two percentiles, our results are robust to lowering the minimum earnings threshold to zero or raising it nontrivially. The chosen threshold ensures as inclusive a sample as possible, while also screening out individuals with idiosyncratic labor market attachment or severely misreported earnings.

FTFY—defined here as being employed at least 50 weeks out of the prior year and working 40 or more usual hours per week. This subsample contains approximately 2.87 million individuals.

We measure pay as the sum of wage-and-salary and self-employment earnings over the prior year. We convert earnings observations to 2021 dollars using the Bureau of Labor Statistics' CPI-Urban series.[3]

II. The Gender Pay Gap is Large in the Tails and Small in the Middle

We begin by estimating the gender log earnings gap across the earnings distribution in two-percentile-wide bins. Specifically, for each $p \in \{.02, .04, \ldots, 1\}$, we construct the estimate $\hat{g}_p = \hat{E}[y_i^w \mid \lceil 50 * q_i^w \rceil / 50 = p] - \hat{E}[y_i^m \mid \lceil 50 * q_i^m \rceil / 50 = p]$, where y denotes log earnings, the superscript denotes individual i's gender (w for woman, m for man), and q_i^s is the quantile rank of i, who is of gender s, in the s-specific earnings distribution. Figure 1 plots the estimates for both full and FTFY samples, centering the bins by subtracting .01 from each percentile group p.[4]

The gender gap estimates for the FTFY sample follow the familiar glass-ceiling pattern that has arisen in other U.S. and European datasets: the gap is small and relatively constant at around 15 log points in the first 30 percentiles of the earnings distribution, and then rises (women's relative earnings fall) thereafter. The decline in FTFY women's relative earnings is steep toward the top of the earnings distribution: the gap at the 85th percentile is 23.0 log points, while the gap at the 99th percentile is 44.2 log points (SE of difference: 0.32 log points).

The pattern looks different, however, when we consider the full sample of workers. While the glass-ceiling dynamic remains, there is now a stronger "sticky-floor" dynamic: the gap is extremely large at the bottom of the distribution, and then declines steadily between the fifth and

[3] The ACS is fielded continuously over the course of the calendar year, asking respondents about income earned over the 12 months immediately prior to the survey week.

[4] Because the minimum earnings threshold artificially constrains the gap toward zero at the very bottom, we omit the bottom percentile groups .01 and .03 from the graph. The pattern is robust to including the .03 percentile group.

30th percentiles. The gap at the fifth percentile is 55.7 log points, which exceeds the gap at the 99th percentile by 5.5 log points. (SE of difference: 0.28 log points.)

III. The Gender Pay Gap Varies by Education Level—especially in the Tails

What are the implications of these tail dynamics for variation in the gender gap across different skill positions in the labor market? We are aware of little research that has analyzed this question.[5] Our companion paper, Binder et al. (2023), makes progress in this direction by linking a sample of recent postsecondary graduates' transcript records to the ACS. Here, we present a briefer overview of the gender gap across a coarser skill distribution—but including individuals who did not complete postsecondary degrees.

Figure 2 divides our sample into six education statuses—less than high school degree, high school degree only, some college without a two- or four-year degree, associate degree, bachelor's degree, advanced degree—and plots selected gender gap statistics across each of these.

As shown in Panel A, several patterns emerge. First, the mean gender pay gap is fairly constant across education statuses: it is around 47 log points among high school dropouts, and 40-44 log points for all other education statuses. However, the mean hides heterogeneous effects at the tails, which we estimate via conditional quantile regression. The gap at the 15th percentile is large among high school dropouts at 59.9 log points, while it is 38.7 log points for advanced degree holders (SE of difference: 0.36 log points). On the other hand, the gap at the 98th percentile is 38.2 log points among high school dropouts, but 56.3 log points among bachelor's and 66.7 log points among advanced degree holders (SE of difference: 0.36 and 0.31 log points, respectively).

[5] Brown and Corcoran (1997) is an important exception, but their study applies to an earlier time period. Binder et al. (2023) compare their findings to those of Brown and Corcoran's paper.

Panel B of Figure 2 shows the same set of statistics for the FTFY sample. The mean gender gap remains constant across the education distribution. Bottom-level inequality remains uniquely low, and top-level inequality uniquely high, among the most-educated workers. However, bottom-level inequality no longer rises as substantially as education attainment falls. Including non-FTFY workers in the sample thus affects the extent to which we observe sticky-floor effects.

Table 1 shows gender differences in labor supply and their contributions to the average pay gap for each education status. The gender gap in the share working full-year (at least 50 weeks out of the year) is small and does not vary across education statuses. The gender gap in the share working full-year and at least 50 hours per week is also relatively constant. On the other hand, the most varied labor supply gap occurs at the FTFY margin itself, ranging from 20.9 percentage points among high school dropouts to 13.7 percentage points among advanced degree holders (SE of difference: 0.32 percentage points). The last column of the table quantifies the contribution of labor supply differences to the mean gender pay gap.[6,7] Such differences widen the mean gender pay gap among those without a high school degree by 23.6 log points, compared with a widening of only 16.2 log points among advanced degree holders (SE of difference: 0.72 percentage points). Labor supply effects are smaller and less heterogeneous in the FTFY sample, widening the mean pay gap by only 3.2 to 4.5 log points.

These findings expand on the hypothesis regarding intensive work hours and the gender gap in top professional occupations (e.g Goldin 2014). While this theory has been shown to be

[6] We allow labor supply to affect earnings in a flexible manner by including one dummy variable for each of 54 different labor supply statuses. These variables capture the interaction of six "weeks-worked" statuses (1-13, 14-26, 27-39, 40-47, 48-49, 50-52) with nine "usual-weekly-hours" statuses (1-20, 21-30, 31-39, 40, 41-49, 50-54, 55-64, 65-74, 75+). For the FTFY subsample, labor supply variables consist solely of dummies for each of the six usual-weekly-hours statuses above 39 hours.

[7] Within each education status, we first regress earnings on a female dummy to get a baseline mean pay gap estimate. We then repeat the regression with the rich set of labor supply controls. The difference in the female dummy coefficient between the two specifications equals the contribution of labor supply to the mean pay gap.

important in certain high-earning sectors, it appears that variation in weekly hours up to but not exceeding 40—especially among individuals without advanced degrees—is the primary margin by which labor supply affects the gender pay gap at the population level.

IV. The Gap has Declined since 2006, Especially among the Most-Educated

We conclude with a brief overview of change in the mean gender pay gap over time. We construct an analogous sample from the first three years of the ACS: 2005-07, to which we refer as "2006." As shown in Panel A of Figure 3, the full-sample mean gap declined by 11.1 log points, or roughly 25%, between our two sample periods. This is a substantive decline that challenges the perception of stagnating progress toward gender equality. The mean pay gap among FTFY workers declined by 6.4 log points (not shown for the sake of brevity). However, this decline is also approximately 25% of the baseline pay gap observed in 2006.

Panel A shows an economically significant decline in the mean pay gap at all education statuses. The observed decline is largest among bachelor's and advanced degree holders. These groups experienced a 10.9 and an 11.1-log-point decline, respectively: similar to the decline observed in the full sample, even though part of the full-sample decline is driven by women's education attainment increasing by more than men's during this time. While average gender inequality used to be as high at the top of the education distribution as at the bottom, it is now much less distinctive at the top.

Panel B suggests that labor supply changes constitute a proximate mechanism behind the observed pattern of pay gap changes. It shows the gender gap in the share of individuals working FTFY by education and time. Just as for pay, we see the largest improvement over time in women's

relative labor supply among bachelor's and advanced degree holders.[8] There has been comparatively much less change in women's relative labor supply among groups of individuals without postsecondary degrees. Thus, while bachelor's and advanced degree-holding women experienced the largest FTFY employment gaps with their male counterparts in 2006, they experienced the smallest gaps in the most-recent period.

To provide a sense of the relationship between labor supply and pay changes, for each education status, we regress pay on a FTFY dummy[9] and multiply the FTFY coefficient by the observed change in the FTFY gap between the two time periods. This simple method finds that FTFY changes predict a decline in the full-sample mean pay gap of 4.2 log points, or 37.8% of the observed decline. Among bachelor's degree holders, FTFY changes predict a mean-pay-gap decline of 6.9 log points, or 63.3% of the observed decline; and among advanced degree holders, FTFY changes account for 10.5 log points, or 94.6%, of the observed decline. In contrast, FTFY changes predict small changes in the mean pay gap among those without college training.

In our most-recent sample, 55.4% of prime-working-age women did not have a bachelor's degree and 45.1% had less than two years of college education. These groups now experience at least as high gender-based inequality in pay and labor supply as do their college-graduate counterparts. We suspect the causes of this inequality likely differ across education groups, and provide some evidence to this point in the related sample we study in Binder et al. (2023). Going forward, researchers should place more emphasis on the goal of understanding the mechanisms driving gender inequality among workers without bachelor's degrees.

[8] Similarly, Binder (2021) found that the FTFY employment rate for college-educated mothers grew by much more than did the rate for less-educated mothers between 1999 and 2016.

[9] We use the most-recent sample to estimate these coefficients, and control for gender in the estimation. Gender-specific coefficients differ from the pooled coefficient used above by no more than a few log points in all cases.

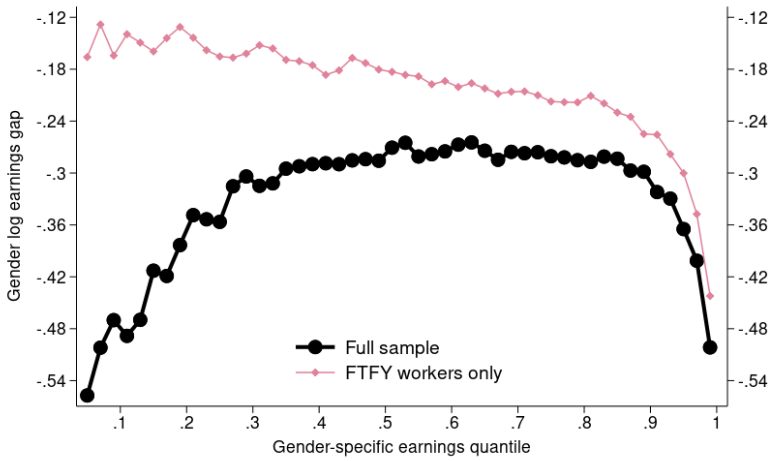


FIGURE 1. THE GENDER PAY GAP AT EACH PERCENTILE OF THE EARNINGS DISTRIBUTION

Source: 2018, 2019, and 2022 American Community Surveys.
Notes: See Section I for further description of sample. Pay is measured as annual earnings throughout this paper. All point estimates are statistically significant at the 1 percent level.

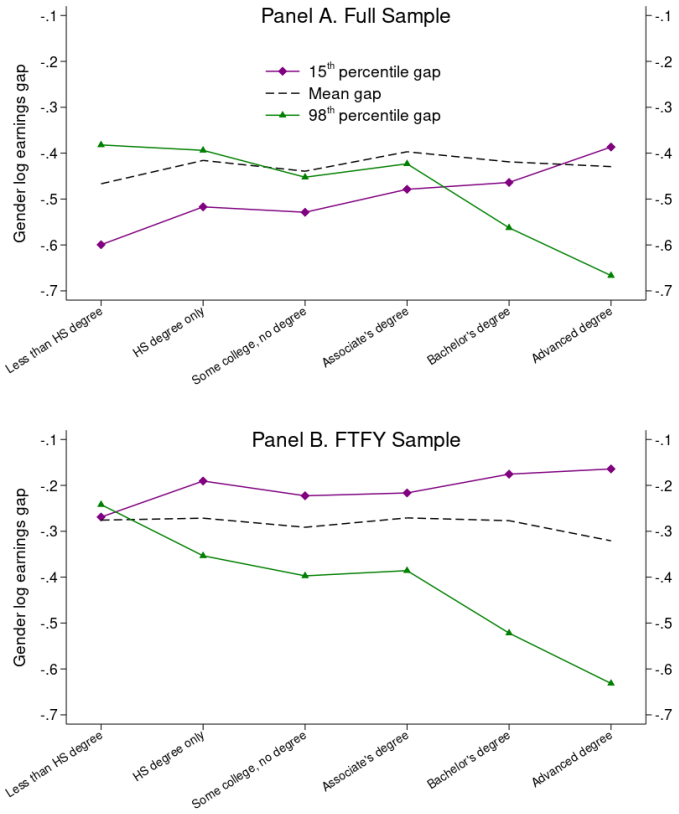


FIGURE 2. 15TH, MEAN, AND 98TH PERCENTILE GENDER PAY GAPS BY EDUCATION STATUS

Source: 2018, 2019, and 2022 American Community Surveys.
Note: See Section I for description of sample. All point estimates are statistically significant at the 1 percent level.

TABLE 1. AVERAGE LABOR SUPPLIES AND THEIR CONTRIBUTIONS TO THE GENDER PAY GAP

Subgroup	Share employed…	Women	Men	Diff	Mean gender pay gap…	
Less than high school degree	Full-year	0.750	0.802	-0.052	in full sample	-0.467
	…and 40+ wkly hrs	0.488	0.697	-0.209	…explained by lab sup	-0.236
	…and 50+ wkly hrs	0.062	0.161	-0.099	in FTFY sample	-0.276
					…explained by lab sup	-0.032
High school graduates	Full-year	0.814	0.847	-0.033	in full sample	-0.416
	…and 40+ wkly hrs	0.583	0.754	-0.171	…explained by lab sup	-0.212
	…and 50+ wkly hrs	0.069	0.190	-0.121	in FTFY sample	-0.271
					…explained by lab sup	-0.042
Some college, no degree	Full-year	0.817	0.863	-0.046	in full sample	-0.439
	…and 40+ wkly hrs	0.609	0.776	-0.167	…explained by lab sup	-0.214
	…and 50+ wkly hrs	0.088	0.225	-0.137	in FTFY sample	-0.291
					…explained by lab sup	-0.045
Associate's degree	Full-year	0.844	0.891	-0.047	in full sample	-0.397
	…and 40+ wkly hrs	0.618	0.810	-0.192	…explained by lab sup	-0.196
	…and 50+ wkly hrs	0.086	0.222	-0.136	in FTFY sample	-0.271
					…explained by lab sup	-0.038
Bachelor's degree	Full-year	0.841	0.899	-0.058	in full sample	-0.419
	…and 40+ wkly hrs	0.677	0.835	-0.158	…explained by lab sup	-0.201
	…and 50+ wkly hrs	0.127	0.242	-0.115	in FTFY sample	-0.277
					…explained by lab sup	-0.036
Advanced degree	Full-year	0.842	0.903	-0.061	in full sample	-0.429
	…and 40+ wkly hrs	0.713	0.850	-0.137	…explained by lab sup	-0.162
	…and 50+ wkly hrs	0.178	0.296	-0.118	in FTFY sample	-0.321
					…explained by lab sup	-0.034

Source: 2018, 2019, and 2022 American Community Surveys.
Notes: See Section I for description of sample. The last column reports the mean gender log earnings gap and the portion of the gap that is explained by controlling for labor supply. See Section III for description of methodology. All point estimates are statistically significant at the 1 percent level.

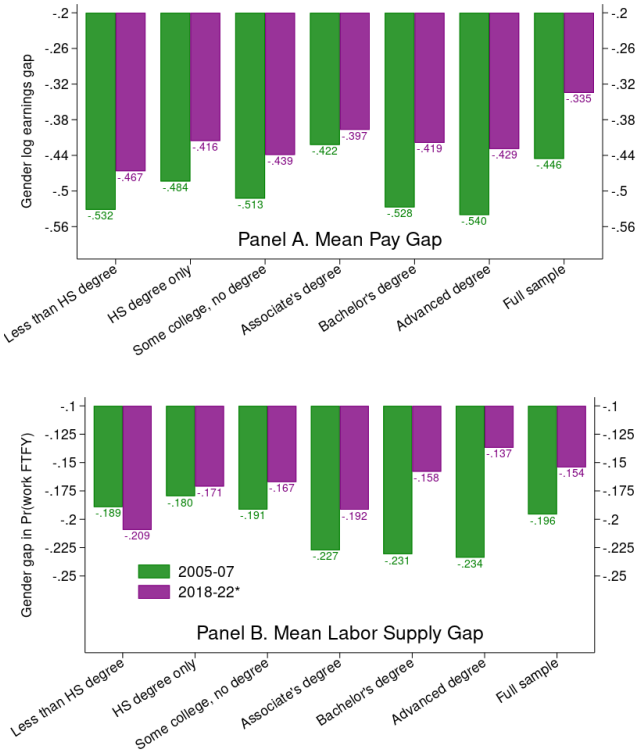


FIGURE 3. COMPARISON OF GENDER PAY AND LABOR SUPPLY GAPS BETWEEN 2005-2007 AND 2018-2022*

Source: 2005-2007; 2018, 2019, and 2022 (referred to as "2018-22*") American Community Surveys.
Note: See Data section or notes to Figure 1 for description of sample. All point estimates are statistically significant at the 1 percent level.

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